Let A be an array of n distinct numbers. We say that an index $1 \leq i \leq n$ is "prefix-maximum" if $A[i]$ is the biggest number so far, that is, if $A[j] < A[i]$ for all $j < i$.
Let $\mathrm{pf}(A)$ denote the number of prefix-maximum indices of A. What is $\mathrm{pf}(A)$ if A is sorted (increasing)?

Options :

6406533484103. ✖ 0

6406533484104. ✖ 1

6406533484105. ✖ $n-1$

6406533484106. ✔ n

Question Number : 137 Question Id : 6406531030006 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose that we permute the elements of A uniformly at random to get an array B. What is $\mathbb{E}[\mathrm{pf}(B)]$?

Options :

6406533484107. ✖ 0

6406533484108. ✖ 1

6406533484109. ✖ $n/2$

6406533484110. ✔ $1 + 1/2 + 1/3 + \cdots + 1/n$

LLM

Section Id :	64065374262
Section Number :	8
Section type :	Online

Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653157130
Question Shuffling Allowed :	No

Question Number : 138 Question Id : 6406531030007 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533484111. ✔ YES

6406533484112. ✖ NO

Question Number : 139 Question Id : 6406531030008 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

General Note:

• If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.

• In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

Options :

6406533484113. ✔ Instructions has been mentioned above.

6406533484114. ✖ This Instructions is just for a reference & not for an evaluation.

Sub-Section Number : 2

Sub-Section Id : 640653157131

Question Shuffling Allowed : No

Question Id : 6406531030009 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (140 to 145)

Question Label : Comprehension

Consider the following dictionary with the number of word occurrences in a corpus:

```
wo =  { "deeper": 5,
        "keener": 6,
        "sweeter": 7,}
```

d e p er k n s we t <w>

Note: Append identifier/special symbol </w> to each word at the end.

You will be learning a byte pair encoding, answer the given subquestions in that context:

Sub questions

Question Number : 140 Question Id : 6406531030010 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many tokens are there in the initial vocabulary?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 141 Question Id : 6406531030011 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following pairs has the least frequency before any merge?

Options :

6406533484116. ✖ ('r', '</w>') 18

6406533484117. ✖ ('w', 'e') 7

6406533484118. ✔ ('p', 'e') 5

6406533484119. ✖ ('e', 'n') 6

6406533484120. ✖ ('e', 'r') 18

6406533484121. ✖ None of these.

Question Number : 142 Question Id : 6406531030012 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the most frequent byte-pair before the very first merge? Say the most frequent byte pair is ('a','b'), then enter **"ab"** (without quotes and white spaces). If there is a tie between two or more candidates, pick the one that occurs first in the original vocabulary.

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

ee

Question Number : 143 Question Id : 6406531030013 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the frequency of the most frequent byte-pair before the very first merge?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

18

Question Number : 144 Question Id : 6406531030014 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many tokens in the vocabulary will have their frequency revised/reduced after first merge?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 145 Question Id : 6406531030015 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

After the first merge, how many tokens are there in the updated vocabulary?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers : 10 +1

11

Sub-Section Number :	3
Sub-Section Id :	640653157132
Question Shuffling Allowed :	No

Question Id : 6406531030016 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (146 to 147)

Question Label : Comprehension

Consider the following dictionary with the number of word occurrences in a corpus:

```
wo = {'simulation</w>': 4,
      'suspicious</w>': 3,
      'stringent</w>': 4,}
```

Note: Identifier/special symbol </w> is already appended.

Suppose the process of building the vocabulary is stopped after 4 merges with BPE tokenization . The following is the list of first 4 merges in that order:

Merge No.	Token 1	Token 2	Merged
1	'i'	'o'	'io'
2	'u'	's'	'us'
3	's'	'i'	'si'
4	'si'	'm'	'sim'

Table 1: First 4 merges

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531030017 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

In how many tokens will the word 'simultaneous</w>' be tokenized? If this word can not be tokenized with the given vocabulary, then enter -1.

Note: Identifier/special symbol </w> is already appended.

1 sim 2 u 3 l 4 t
5 a 6 n 7 e 8 o
9 us <w> 10

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 147 Question Id : 6406531030018 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

In how many tokens will the word 'stereotypical</w>' be tokenized? If this word can not be tokenized with the given vocabulary, then enter -1.

Note: Identifier/special symbol </w> is already appended.

Since letter y is missing in the vocab

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-1

Sub-Section Number :	4
Sub-Section Id :	640653157133
Question Shuffling Allowed :	Yes

Question Number : 148 Question Id : 6406531030019 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following represent(s) the normalization step(s) in building a tokenizer for the English language?

Options :

6406533484128. ✔ removing accents from characters.

6406533484129. ✔ converting upper case letters to small case letters.

6406533484130. ✖ assigning integers to tokens.

6406533484131. ✖ splitting a sentence by whitespaces.

6406533484132. ✖ the final post-processing step where the tokenizer adds the special tokens.

6406533484133. ✖ none of these.

Question Number : 149 Question Id : 6406531030022 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Choose all the aspects of the pre-training datasets that impact the model's performance.

Options :

6406533484143. ✔ Quality

6406533484144. ✔ Size

6406533484145. ✔ Diversity

6406533484146. ✖ None of these

Sub-Section Number : 5

Sub-Section Id : 640653157134

Question Shuffling Allowed : Yes

Question Number : 150 Question Id : 6406531030020 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

The strikeout words in the passage given below denote the words to be dropped from the original sentence.

"Metacognition ~~is an awareness~~ of one's thought processes and an understanding of the patterns behind them. The term comes from the root word meta, meaning "beyond", or "on top of". Metacognition ~~can take many forms~~, such as reflecting on one's ways of thinking, and knowing when and how oneself and others use ~~particular strategies~~ for problem-solving. There are generally two components of metacognition: cognitive conceptions ~~and cognitive regulation~~ system."

Which of the following represents the correct target sequence, with sentinel tokens, to the baseline model (according to T5 study) that uses the pre-training denoising objective? The characters inside the square brackets are the sentinel tokens and [z] represents the end of the sentinel token in a sentence

Options :

6406533484134. ✖ [v] is an awareness [w] can take many forms [x] particular strategies [y] and cognitive regulation

6406533484135. ✔ [v] is an awareness [w] can take many forms [x] particular strategies [y] and cognitive regulation [z]

6406533484136. ✖ [v] is an awareness [w] can take many forms [x] and cognitive regulation [y] particular strategies

6406533484137. ✖ [v] is an awareness [w] particular strategies [x] can take many forms [y] and cognitive regulation

6406533484138. ✖ [a] is an awareness [b] can take many forms [c] particular strategies [d] and cognitive regulation

6406533484139. ✔ [a] is an awareness [b] can take many forms [c] particular strategies [d] and cognitive regulation[z]

6406533484140. ✖ [a] is an awareness [b] can take many forms [c] and cognitive regulation

6406533484141. ✖ None of these.

Sub-Section Number : 6

Sub-Section Id : 640653157135

Question Shuffling Allowed : Yes

Question Number : 151 Question Id : 6406531030021 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose you are working on prefix language modeling. The sequence length is 32 and the first two tokens represent the task specific prefix. How many non-infinity elements are there in the mask for computing attention scores?

$32 - 2 = 30$

$\frac{\overset{15}{\cancel{30}} \times 31}{\cancel{2}} = 465 + 64 = 529$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

529

Sub-Section Number : 7

Sub-Section Id : 640653157136

Question Shuffling Allowed : Yes

Question Number : 152 Question Id : 6406531030023 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following components in the data pre-processing pipeline removes pages that contain bad words?

Options :

6406533484147. ✖ Language Identification

6406533484148. ✖ Exact Deduplication

6406533484149. ✖ Fuzzy Deduplication

6406533484150. ✖ ML classifiers for quality filtering

6406533484151. ✔ Simple heuristics to detect toxic contents

Sub-Section Number : 8

Sub-Section Id : 640653157137

Question Shuffling Allowed : Yes

Question Number : 153 Question Id : 6406531030024 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is an important implication of the scaling law:

Options :

6406533484152. ✖ Increasing the size of the model only can improve test performance, irrespective of the size of the training data.

6406533484153. ✖ Increasing the size of the data only can improve test performance, irrespective of the size of the model.

6406533484154. ✔ Both the size of the model and that of the data, have to be increased appropriately to improve test performance.

6406533484155. ✖ None of these.

Sub-Section Number : 9

Sub-Section Id : 640653157138

Question Shuffling Allowed : Yes

Question Number : 154 Question Id : 6406531030025 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are the design choices for building a large language model?

Options :

6406533484156. ✔ Activation function in feed forward layer

6406533484157. ✔ Training dataset.

6406533484158. ✔ Positional encoding

6406533484159. ✔ Attention mechanism.

6406533484160. ✖ None of these.

Question Number : 155 Question Id : 6406531030026 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider an ideal data preprocessing pipeline to prepare a dataset. Which of the following sentences or sets of paragraphs will NOT pass ~~through the pipeline as it is?~~

Options :

6406533484161. ✔ “You idiot. All those people from the country Sumatrazia are taking our jobs. If I get a chance, I will wipe that country off of the face of the planet.”

6406533484162. ✔ Rajesh is an amazing businessman. He holds a savings account with Dena bank. His phone number is 9876543210. PII

6406533484163. ✔ Deep learning is a subset of Machine Learning.
Deep learning is a subset of Machine Learning.

6406533484164. ✖ India’s space agency ISRO says it has successfully brought back into Earth’s orbit a part of the rocket that carried its historic Moon mission recently.

6406533484165. ✖ Last Saturday, the Board of Control for Cricket in India (BCCI) officially communicated to the International Cricket Council (ICC) its decision to not send a team for the 2025 ICC Champions Trophy to be held in Pakistan from February 19 to March 9.

6406533484166. ✖ None of these.

i-NLP